



Personal Reflection on Performance Analysis and CNN Validation

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Before completing this laboratory, I understood that training accuracy and validation accuracy were important measures of a neural network, but I was less confident about interpreting them together. It was easy to see a high accuracy value and assume that the model was successful. This laboratory helped me recognize that performance analysis requires examining training behavior, validation behavior, and final test results as parts of the same process.

My contribution focused on performance analysis and architectural validation. I examined training trends for possible signs of overfitting or underfitting and considered whether the results were consistent with the architecture being used. This made training more meaningful to me: I was interested not only in whether the code ran, but also in what its results revealed about how the network was learning.

One of the most useful lessons was that high training accuracy does not automatically mean a model will generalize well to unseen data. Comparing training and validation performance provides a more complete picture. If training performance keeps improving while validation performance stops improving or worsens, the model may be fitting its training data too closely. Looking for this pattern helped me understand why validation data matters in deep learning.

I also helped validate the workflow from data loading through final evaluation. Following the complete process showed me how its components depend on one another: preprocessing prepares the images and labels; the CNN extracts features; the optimizer updates model parameters; and evaluation provides evidence about performance on data not used to fit the model. This reinforced that performance analysis cannot be separated from the architectural and training decisions behind the results.

Examining the model architecture also gave me a clearer way to interpret its performance. The network used two convolutional layers and two max-pooling layers before flattening the extracted features and passing them to a final `Dense` layer. Seeing the progression from feature extraction to classification reminded me that an accuracy value is the outcome of a learning process, not an explanation by itself.

A challenge was learning to distinguish a result that looks impressive from one that is supported by the available evidence. MNIST is a relatively controlled task, so strong performance on it does not establish that the same architecture would perform equally well on a harder image dataset. This helped me think more carefully about both what a metric tells us and what it cannot tell us. The laboratory also showed me why experimentation matters. Changing the batch size or number of epochs can affect training time and the course of training. The notebook comparisons showed that different settings can produce similar validation accuracy. Choosing parameters therefore involves more than maximizing one number; it also means considering the cost of training and whether the performance difference is meaningful.

My main takeaway is that performance analysis is an interpretive part of machine learning. Metrics need to be read in context rather than accepted without question. I now have a stronger foundation for reading training results, identifying possible generalization problems, and

explaining why a model performs as it does. I can apply that approach as I work with larger and more complex neural-network architectures.